

L1.9 L'Hôpital's Rule

Recall: L1.8 Linearization & Taylor

Name _____

Solo – 9 minutes

1. Build the linearization of $f(x) = \sqrt[3]{x}$ at $a = 8$, then use it to estimate $\sqrt[3]{8.5}$.

2. A function has these values at $x = 0$:

n	0	1	2	3
$f^{(n)}(0)$	1	-2	6	-12

Write $T_3(x)$.

3. You want to estimate $\sqrt{101}$. What center a would you use, and why that one?